

EXPERIMENT 1

DDL Commands – CREATE, ALTER, DROP

1) Create a table name STUDENT with following structure.

#	Column Name	Description	Data Type
1	RegNo	Registration Number	NUMBER(3)
2	Name	Student Name	VARCHAR(15)
3	Gender	Gender of the student	CHAR(1)
4	DOB	Date of Birth	DATE
5	MobileNo	Mobile Number	NUMBER(10)
6	City	Location of stay	VARCHAR(15)

```
mysql> create table student(Regno int(3),Name char(15),gender char(1),Dob int(10),mobileno int(10),city char(10));
Query OK, 0 rows affected (0.14 sec)

mysql> desc student;
+-----+-----+-----+-----+-----+-----+
| Field | Type  | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| Regno | int(3) | YES  |     | NULL    |       |
| Name  | char(15) | YES  |     | NULL    |       |
| gender | char(1) | YES  |     | NULL    |       |
| Dob   | int(10) | YES  |     | NULL    |       |
| mobileno | int(10) | YES  |     | NULL    |       |
| city  | char(10) | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)
```

2) Create a table name FACULTY with following structure.

#	Column Name	Description	Data Type
1	FacNo	Faculty Identifier	VARCHAR(4)
2	FacName	Faculty Name	VARCHAR(15)
3	Gender	Gender of faculty	CHAR(1)
4	DOB	Date of Birth	DATE
5	DOJ	Date of Join	DATE
6	MobileNo	Mobile Number	NUMBER(10)

```
mysql> create table faculty(Facno int(3),FacName char(15),gender char(1),Dob int(10),mobilen no int(10),DOJ int(10));
Query OK, 0 rows affected (0.06 sec)

mysql> desc faculty;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| Facno | int(3) | YES | | NULL | |
| FacName | char(15) | YES | | NULL | |
| gender | char(1) | YES | | NULL | |
| Dob | int(10) | YES | | NULL | |
| mobilen o | int(10) | YES | | NULL | |
| DOJ | int(10) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)
```

3) Create a table name DEPARTMENT with following structure.

#	Column Name	Description	Data Type
1	DeptNo	Department Identifier	VARCHAR(4)
2	DeptName	Department Name	VARCHAR(15)
3	DeptHead	Department Head	VARCHAR(4)

```
mysql> create table department(deptno int(10),deptname char(10),depthead char(10));
Query OK, 0 rows affected (0.11 sec)

mysql> desc department;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| deptno | int(10) | YES | | NULL | |
| deptname | char(10) | YES | | NULL | |
| depthead | char(10) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

4) Create a table name COURSE with following structure.

#	Column Name	Description	Data Type
1	CourseNo	Course Identifier	VARCHAR(3)
2	CourseDesc	Course Description	VARCHAR(14)
3	CourseType	Course Type	CHAR(1)
4	SemNo	Semester Number	CHAR(1)
5	HallNo	Hall Number	VARCHAR(4)
6	FacNo	Faculty Identifier	VARCHAR(4)

```
mysql> create table course(courseno int(3),coursedesc char(15),coursetype char(1),semno int(10),hallno int(10),Facno int(10));
Query OK, 0 rows affected (0.09 sec)

mysql> desc course;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| courseno | int(3) | YES | | NULL | |
| coursedesc | char(15) | YES | | NULL | |
| coursetype | char(1) | YES | | NULL | |
| semno | int(10) | YES | | NULL | |
| hallno | int(10) | YES | | NULL | |
| Facno | int(10) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)
```

5) Modify the table FACULTY by adding a column name Dept of datatype VARCHAR(10)

```
mysql> alter table faculty add dept char(10);
Query OK, 2 rows affected (0.09 sec)
Records: 2 Duplicates: 0 Warnings: 0

mysql> desc faculty;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| Facno | int(3) | NO | PRI | NULL | |
| FacName | char(15) | YES | | NULL | |
| gender | char(1) | YES | | NULL | |
| Dob | int(10) | YES | | NULL | |
| mobileno | int(10) | YES | | NULL | |
| DOJ | int(10) | YES | | NULL | |
| dept | char(10) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
7 rows in set (0.00 sec)
```

EXPERIMENT 2

DDL Commands with Constraints – PRIMARY, FOREIGN KEY, UNIQUE, CHECK

1) Alter the table STUDENT2 with following structure.

#	Column Name	Constraints
1	RegNo	PRIMARY KEY

```
mysql> Alter table student2 ADD PRIMARY KEY(RegNo);
Query OK, 0 rows affected (0.05 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> desc student2;
```

Field	Type	Null	Key	Default	Extra
RegNo	int	NO	PRI	NULL	
S1Name	varchar(15)	YES		NULL	
Age	char(2)	YES		NULL	
MobileNo	int	YES		NULL	
address	varchar(15)	YES		NULL	

5 rows in set (0.01 sec)

2) Alter the table name FACULTY with following structure.

#	Column Name	Constraints
1	FacNo	PRIMARY KEY
2	Gender	CHECK 'M' or 'F'

```
mysql> alter table faculty add primary key(facno);
Query OK, 0 rows affected (0.07 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> alter table faculty add check(gender='M'or'F');
Query OK, 0 rows affected (0.06 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> desc faculty;
+-----+-----+-----+-----+-----+-----+
| Field      | Type      | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| Facno      | int(3)    | NO   | PRI | NULL    |       |
| FacName    | char(15)  | YES  |     | NULL    |       |
| gender     | char(1)   | YES  |     | NULL    |       |
| Dob        | int(10)   | YES  |     | NULL    |       |
| mobileno   | int(10)   | YES  |     | NULL    |       |
| DOJ        | int(10)   | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)
```

3)Alter the table name DEPARTMENT with following structure.

#	Column Name	Constraints
1	DeptNo	PRIMARY KEY

```
mysql> alter table department add primary key(deptno);
Query OK, 0 rows affected (0.11 sec)

mysql> desc department;
+-----+-----+-----+-----+-----+-----+
| Field      | Type      | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| deptno     | int(10)   | NO   | PRI | NULL    |       |
| deptname   | char(10)  | YES  |     | NULL    |       |
| depthead   | char(10)  | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

4)Alter the table name COURSE with following structure.

#	Column Name	Constraints
1	CourseNo	PRIMARY KEY
2	SemNo	1 to 6


```
mysql> alter table course add primary key(courseno);
Query OK, 0 rows affected (0.07 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> alter table course add check(semno>=1&&semno<=6);
Query OK, 0 rows affected (0.11 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> desc course;
+-----+-----+-----+-----+-----+-----+
| Field      | Type      | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| courseno   | int(3)    | NO   | PRI | NULL    |       |
| coursedesc | char(15)  | YES  |     | NULL    |       |
| coursetype | char(1)   | YES  |     | NULL    |       |
| semno      | int(10)   | YES  |     | NULL    |       |
| hallno     | int(10)   | YES  |     | NULL    |       |
| Facno      | int(10)   | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)
```

5) Alter the table name STUDENT2 with following structure.

#	Column Name	Constraints
1	S1Name	UNIQUE KEY

```
mysql> Alter table student2 ADD UNIQUE KEY(S1Name);
Query OK, 0 rows affected (0.07 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> desc student2;
+-----+-----+-----+-----+-----+-----+
| Field      | Type      | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| RegNo      | int       | NO   | PRI | NULL    |       |
| S1Name     | varchar(15) | YES  | UNI | NULL    |       |
| Age        | char(2)   | YES  |     | NULL    |       |
| MobileNo   | int       | YES  |     | NULL    |       |
| address    | varchar(15) | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

6) After the STUDENT2 & STUDENT3 tables are successfully created, test if you can add a constraint FOREIGN KEY to the RegNo of this table

```
mysql> ALTER TABLE student1 modify address int not null;  
Query OK, 0 rows affected (0.07 sec)  
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> desc student1;
```

Field	Type	Null	Key	Default	Extra
RegNo	int	NO	PRI	NULL	
S1Name	varchar(15)	YES		NULL	
Age	char(2)	YES	UNI	NULL	
DOB	int	YES		NULL	
address	int	NO		NULL	

```
5 rows in set (0.00 sec)
```

EXPERIMENT -3

DML Commands – INSERT, SELECT

1. Populate all the five tables with your own data.

```
mysql> insert into student values
-> (19221101,'Jayashree','F','2005-06-15','9876543210','Chennai'),
-> (19221102,'Priya','F','2005-09-20','9876543211','Coimbatore'),
-> (19221103,'Kavin','M','2004-12-10','9876543212','Madurai'),
-> (19221104,'Arun','M','2005-03-18','9876543213','Salem'),
-> (19221105,'Divya','F','2005-11-25','9876543214','Trichy');
Query OK, 5 rows affected (0.02 sec)
Records: 5 Duplicates: 0 Warnings: 0

mysql> select * from student;
+-----+-----+-----+-----+-----+-----+
| Reg_no | Name   | Gender | DOB       | Mobile_no | City      |
+-----+-----+-----+-----+-----+-----+
| 19221101 | Jayashree | F      | 2005-06-15 | 9876543210 | Chennai  |
| 19221102 | Priya    | F      | 2005-09-20 | 9876543211 | Coimbatore |
| 19221103 | Kavin    | M      | 2004-12-10 | 9876543212 | Madurai   |
| 19221104 | Arun     | M      | 2005-03-18 | 9876543213 | Salem    |
| 19221105 | Divya    | F      | 2005-11-25 | 9876543214 | Trichy    |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql>
```

```
mysql> insert into faculty values
-> ('F001','Meena','F','1985-04-10','2010-06-12','9876500001','CSE','N'),
-> ('F002','Ramesh','M','1980-08-15','2008-05-20','9876500002','ECE','Y'),
-> ('F003','Suresh','M','1983-12-18','2012-07-25','9876500003','EEE','N'),
-> ('F004','Anitha','F','1988-03-14','2015-09-10','9876500004','IT','Y'),
-> ('F005','Kumar','M','1982-11-28','2009-01-05','9876500005','MECH','N');
Query OK, 5 rows affected (0.02 sec)
Records: 5 Duplicates: 0 Warnings: 0

mysql> select * from faculty;
+-----+-----+-----+-----+-----+-----+-----+
| Facno | Facname | Gender | DOB       | DOJ       | Mobileno | Deptno | Resigned |
+-----+-----+-----+-----+-----+-----+-----+
| F001  | Meena   | F      | 1985-04-10 | 2010-06-12 | 9876500001 | CSE    | N        |
| F002  | Ramesh  | M      | 1980-08-15 | 2008-05-20 | 9876500002 | ECE    | Y        |
| F003  | Suresh  | M      | 1983-12-18 | 2012-07-25 | 9876500003 | EEE    | N        |
| F004  | Anitha  | F      | 1988-03-14 | 2015-09-10 | 9876500004 | IT     | Y        |
| F005  | Kumar   | M      | 1982-11-28 | 2009-01-05 | 9876500005 | MECH   | N        |
+-----+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql>
```



```
mysql> insert into course values
-> ('CS101','Programming in C','Mandatory',1,'H001','F001'),
-> ('CS102','Database Management','Mandatory',2,'H002','F002'),
-> ('CS103','Data Structures','Mandatory',3,'H003','F003'),
-> ('CS104','Computer Networks','Elective',4,'H004','F004'),
-> ('CS105','Artificial Intelligence','Elective',5,'H005','F005');
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0

mysql> select * from course;
+-----+-----+-----+-----+-----+-----+
| CourseNo | CourseDesc | CourseType | SemNo | HallNo | FacNo |
+-----+-----+-----+-----+-----+-----+
| CS101 | Programming in C | Mandatory | 1 | H001 | F001 |
| CS102 | Database Management | Mandatory | 2 | H002 | F002 |
| CS103 | Data Structures | Mandatory | 3 | H003 | F003 |
| CS104 | Computer Networks | Elective | 4 | H004 | F004 |
| CS105 | Artificial Intelligence | Elective | 5 | H005 | F005 |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql>
```

```
mysql> insert into department values
-> (1,'Computer Science','Dr.Meena'),
-> (2,'Electronics','Dr.Ramesh'),
-> (3,'Electrical','Dr.Suresh'),
-> (4,'Information Technology','Dr.Anitha'),
-> (5,'Mechanical','Dr.Kumar');
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0

mysql> select * from department;
+-----+-----+-----+
| Deptno | Deptname | Depthead |
+-----+-----+-----+
| 1 | Computer Science | Dr.Meena |
| 2 | Electronics | Dr.Ramesh |
| 3 | Electrical | Dr.Suresh |
| 4 | Information Technology | Dr.Anitha |
| 5 | Mechanical | Dr.Kumar |
+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> _
```

```
mysql> insert into hall values
-> ('H001','Block A',60),
-> ('H002','Block B',55),
-> ('H003','Block C',50),
-> ('H004','Block D',45),
-> ('H005','Block E',70);
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0

mysql> select * from hall;
+-----+-----+-----+
| HallNo | Hallname | Capacity |
+-----+-----+-----+
| H001 | Block A | 60 |
| H002 | Block B | 55 |
| H003 | Block C | 50 |
| H004 | Block D | 45 |
| H005 | Block E | 70 |
+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> _
```

2) View all the records from the five tables.

```
mysql> select * from student;
```

Reg_no	Name	Gender	DOB	Mobile_no	City
19221101	Jayashree	F	2005-06-15	9876543210	Chennai
19221102	Priya	F	2005-09-20	9876543211	Coimbatore
19221103	Kavin	M	2004-12-10	9876543212	Madurai
19221104	Arun	M	2005-03-18	9876543213	Salem
19221105	Divya	F	2005-11-25	9876543214	Trichy

5 rows in set (0.00 sec)

```
mysql> _
```

```
mysql> select * from course;
```

Courseno	Coursedes	Coursetype	Semno	Hallno	Facno
CS101	Programming in C	Mandatory	1	H001	F001
CS102	Database Management	Mandatory	2	H002	F002
CS103	Data Structures	Mandatory	3	H003	F003
CS104	Computer Networks	Elective	4	H004	F004
CS105	Artificial Intelligence	Elective	5	H005	F005

5 rows in set (0.00 sec)

```
mysql> _
```

```
mysql> select * from department;
```

Deptno	Deptname	Depthead
1	Computer Science	Dr.Meena
2	Electronics	Dr.Ramesh
3	Electrical	Dr.Suresh
4	Information Technology	Dr.Anitha
5	Mechanical	Dr.Kumar

```
5 rows in set (0.00 sec)

mysql> _
```

```
mysql> select * from hall;
```

Hallno	Hallname	Capacity
H001	Block A	60
H002	Block B	55
H003	Block C	50
H004	Block D	45
H005	Block E	70

```
5 rows in set (0.00 sec)

mysql> _
```

EXPERIMENT - 4

DML Commands with Constraints –UPDATE, DELETE

Questions:

1. Update the value of student name whose register number is '191711342'

```
mysql> UPDATE student
  -> SET Name = 'Jayashree'
  -> WHERE Reg_no = 192524146;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT * FROM student
  -> WHERE Reg_no = 192524146;
+-----+-----+-----+-----+-----+-----+
| Reg_no | Name      | Gender | DOB       | Mobile_no | City    |
+-----+-----+-----+-----+-----+-----+
| 192524146 | Jayashree | F      | 2007-10-20 | 9876543210 | Chennai |
+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> █
```

2. Delete the record in the table FACULTY, who resigned her job.

```
mysql> delete from faculty where facname='Mohan';
Query OK, 1 row affected (0.02 sec)

mysql> select * from faculty;
+-----+-----+-----+-----+-----+-----+-----+-----+
| facno  | facname  | Gender | DOB       | DOJ       | Mobileno | deptno | resigned |
+-----+-----+-----+-----+-----+-----+-----+-----+
| F1922113 | ramesh   | m      | 2005-03-14 | 2014-12-20 | 998263548 | mnr    | y        |
| F1922115 | san      | m      | 2007-05-17 | 2016-09-20 | 998269245 | hyt    | y        |
| F1922112 | raj      | m      | 2005-02-13 | 2009-06-25 | 90865894  | mgr    | y        |
+-----+-----+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> █
```


3. Modify the age for the faculty whose name is 'mohan' with a value '59'.

```
mysql> select * from faculty;
```

facno	facname	Gender	DOB	DOJ	Mobileno	deptno	resigned	age
F1922113	ramesh	m	2005-03-14	2014-12-20	998263548	mnr	y	45
F1922115	san	m	2007-05-17	2016-09-20	998269245	hyt	y	68
1191151	mohan	m	2004-01-12	2010-05-23	990826973	mtr	y	59
1f1922112	raju	m	2005-02-13	2009-06-25	90865894	mgr	y	45

```
4 rows in set (0.00 sec)
```

```
mysql>
```

4. Remove all faculty who are having over 65 years .

```
mysql> delete from faculty where age > 65;  
Query OK, 1 row affected (0.02 sec)  
Rows matched: 1   Changed: 1   Warnings: 0  
  
mysql> select * from faculty;  
+-----+-----+-----+-----+-----+-----+-----+-----+-----+  
| facno      | facname | Gender | DOB          | DOJ          | Mobileno    | deptno | resigned | age |  
+-----+-----+-----+-----+-----+-----+-----+-----+-----+  
| F1922113   | ramesh  | m       | 2005-03-14   | 2014-12-20   | 998263548   | mnr    | y         | 45 |  
| 1191151    | mohan   | m       | 2004-01-12   | 2010-05-23   | 990826973   | mtr    | y         | 59 |  
| 1f1922112  | rajuu   | m       | 2005-02-13   | 2009-06-25   | 90865894    | mgr    | y         | 45 |  
+-----+-----+-----+-----+-----+-----+-----+-----+-----+  
3 rows in set (0.00 sec)  
  
mysql>
```